

* local susy requires gravity.

→ a constant spinor ϵ is not compatible w the symmetries required in a theory of grav. w/ fermions. $\epsilon \rightarrow \epsilon(x)$

$$S = S_2 + S_{3/2}$$

$$S_2 = \frac{1}{2\kappa^2} \int d^4x \, e \, e^{\alpha\mu} e^{b\nu} R_{\mu\nu\alpha\beta}(w)$$

$$S_{3/2} = -\frac{1}{2\kappa} \int d^4x \, e \, \bar{\psi}_\mu \gamma^{\mu\nu\rho} D_\nu \psi_\rho$$

$$e \equiv \det(e_\mu^\alpha) \quad | \quad D_\nu \psi_\mu = \partial_\nu \psi_\mu + \frac{1}{4} (w_\nu)_{ab} \gamma^{ab} \psi_\mu$$

• Torsion-free spin connection $(w_\nu)_{ab}(e)$

$$w_\mu^{ab}(e) = 2 e^{\nu[a} \partial_{[\mu} e_{\nu]}^{b]} - e^{\nu[a} e^{b]\sigma} e_{\mu\sigma} \partial_\nu e_\sigma^c$$

• $\Gamma_{\nu\rho}^\sigma \psi_\sigma$ is not added to the cov. der. since

$$\Gamma_{\nu\rho}^\sigma \psi_\sigma \gamma^{\mu\nu\rho} = 0$$

* Susy transformations:

$$\delta e_\mu^a = \frac{1}{2} \bar{\epsilon} \gamma^a \psi_\mu$$

$$\delta \psi_\mu = D_\mu \epsilon(x) = \partial_\mu \epsilon(x) + \frac{1}{4} w_{\mu ab} \gamma^{ab} \epsilon(x)$$

ψ_μ : Gauge-field of local susy, so

$$\psi_\mu \rightarrow \psi_\mu + \partial_\mu \epsilon \implies \psi_\mu \rightarrow \psi_\mu + D_\mu \epsilon$$

in a curved space time

$$e_\mu^a e_\nu^b = \delta_{\mu\nu}^{\quad ab}$$

$$\implies (\delta e_\mu^a) e_\nu^b = - e_\mu^a \delta e_\nu^b$$

$$\delta e_\mu^a = - e_\mu^\sigma e_\sigma^a e_\nu^b \delta e_\nu^b$$

$$\delta e_\mu^a = - e_\mu^\sigma e_\sigma^a \left(\frac{1}{2} \bar{\epsilon} \gamma^b \psi_\mu \right)$$

$$= - \frac{1}{2} \gamma^a \psi_\mu$$

$$\delta e_a^\sigma = -\frac{1}{2} \bar{\epsilon} \gamma^\sigma \psi_a$$

$$\Rightarrow \det(e_a^\mu) = e$$

$$\delta e \Rightarrow \log(\det(e_a^\mu)) = \text{tr}(\log(e_a^\mu))$$

$$e^{-1} \delta e = \text{Tr}(e^{-1} \delta e_a^\mu)$$

$$\delta e = e \text{Tr}(e_a^\mu \frac{1}{2} \bar{\epsilon} \gamma^\mu \psi_a)$$

$$= e e_a^\mu \frac{1}{2} \bar{\epsilon} \gamma^\mu \psi_a$$

$$\therefore \delta e = e (\frac{1}{2} \bar{\epsilon} \gamma^\mu \psi_\mu)$$

→ Go to first order formalism w/ torsion

Variation of Pure Gravity:

$$\delta S_2 = \frac{1}{2\kappa^2} \int d^4x \delta(e e^\mu_a e^\nu_b R_{\mu\nu ab}(\omega))$$

$$= \frac{1}{2\kappa^2} \int d^4x (\delta e e^\mu_a e^\nu_b R_{\mu\nu ab} + e \delta e^\mu_a e^\nu_b R_{\mu\nu ab} + e e^\mu_a \delta e^\nu_b R_{\mu\nu ab} + e e^\mu_a e^\nu_b \delta R_{\mu\nu ab})$$

$$= \frac{1}{2\kappa^2} \int d^4x [2e \delta e^\mu_a e^\nu_b R_{\mu\nu ab} + \delta e e^\mu_a e^\nu_b R_{\mu\nu ab} + e e^\mu_a e^\nu_b \delta R_{\mu\nu ab}]$$

Since $\delta R_{\mu\nu ab} \sim \nabla_\mu \omega_{\nu ab} - \nabla_\nu \omega_{\mu ab}$

$$= \frac{1}{2\kappa^2} \int d^4x 2\mathcal{M}^{ac} (\delta e_c^\mu) e^\nu_b R_{\mu\nu ab} e + \dots$$

$$= \frac{1}{2\kappa^2} \int d^4x (-e \frac{1}{2} \bar{\epsilon} \gamma^\sigma \psi_\sigma e^\mu_a e^\nu_b R_{\mu\nu ab}) + e 2\mathcal{M}^{ac} (\frac{1}{2} \bar{\epsilon} \gamma^\mu \psi_c) e^\nu_b R_{\mu\nu ab}$$

Define Ricci by: $R_{\mu\nu} \equiv R_{\mu}{}^\sigma{}_{\nu\sigma}$

(i) $\frac{1}{2} e R \bar{\epsilon} \gamma^\sigma \psi_\sigma = - \bar{\epsilon} \frac{1}{2} e (g_{\mu\nu} R) \gamma^\mu \psi^\nu$

(ii) $e \bar{\epsilon} \gamma^\mu \psi_a e^\nu_b R_{\mu\nu ab}$

$$= e \bar{\epsilon} \gamma^\mu \psi_a R_{\mu}{}^b{}_{ab}$$

$$= e \bar{\epsilon} \gamma^\mu \psi_a R_{\mu a}$$

$$= e \bar{e} \gamma^\mu \psi^\nu e_\nu^\alpha R_{\mu\alpha}$$

$$= e \bar{e} \gamma^\mu \psi^\nu R_{\mu\nu}$$

$$\Rightarrow \delta S_2 = \frac{1}{2\kappa^2} \int d^4x e [R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R] \gamma^\mu \psi^\nu$$

Thus giving us Einstein tensor for pure gravity.

Grauling Action:

$$\delta S_{3/2} = -\frac{1}{2\kappa^2} \int d^4x \delta(e \bar{\psi}_\mu \gamma^{\mu\nu\rho} D_\nu \psi_\rho)$$

$$\delta e \bar{\psi}_\mu \gamma^{\mu\nu\rho} D_\nu \psi_\rho + e (\delta \bar{\psi}_\mu) \gamma^{\mu\nu\rho} D_\nu \psi_\rho + e \bar{\psi}_\mu \gamma^{\mu\nu\rho} \delta(D_\nu \psi_\rho)$$

$$- E = mc^2 \leftarrow$$

$$- E^2 = (mc^2)^2 + (pc)^2$$

$$E = pc$$

$$i\partial_t |\psi\rangle = \hat{H} |\psi\rangle$$

$$i\partial_t \psi = \frac{\hbar}{2m} \nabla^2 \psi + \psi V(x)$$

$$\psi = a |\psi_1\rangle + b |\psi_2\rangle$$